```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
```

```
uploaded = files.upload()
```

Choose Files car.avif

car.avif(image/avif) - 30952 bytes, last modified: 9/10/2026 - 100% done

```
Saving car.avif to car.avif
```

```
filename = list(uploaded.keys())[0]
print("Uploaded file:", filename)
```

```
Uploaded file: car.avif
```

```
image = cv2.imread(filename)

print("Image loaded successfully!")
print("Image shape:", image.shape)
```

```
Image loaded successfully!
Image shape: (384, 512, 3)
```

```
from google.colab.patches import cv2_imshow
```

```
cv2_imshow(image)
```




```
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.imshow(image_rgb)
plt.axis('off')
plt.title("Original Image")
plt.show()
```




```
height, width, channels = image.shape

print("Height:", height)
print("Width:", width)
print("Number of Channels:", channels)
print("Data Type:", image.dtype)
print("Resolution:", width, "x", height)
```

```
Height: 384
Width: 512
Number of Channels: 3
Data Type: uint8
Resolution: 512 x 384
```

```
cv2.imwrite("output_image.jpg", image)
cv2.imwrite("output_image.png", image)

print("Image saved successfully in JPEG and PNG formats.")
```

```
Image saved successfully in JPEG and PNG formats.
```

```
import os

jpeg_size = os.path.getsize("output_image.jpg")
png_size = os.path.getsize("output_image.png")

print("JPEG size:", jpeg_size, "bytes")
print("PNG size:", png_size, "bytes")
```

```
JPEG size: 81147 bytes
PNG size: 324077 bytes
```

```
import os

jpeg_size = os.path.getsize("output_image.jpg")
png_size = os.path.getsize("output_image.png")

print("JPEG size:", jpeg_size, "bytes")
print("PNG size:", png_size, "bytes")
```

```
JPEG size: 81147 bytes
PNG size: 324077 bytes
```

```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

plt.imshow(gray, cmap='gray')
plt.axis('off')
plt.title("Grayscale Image")
plt.show()
```




```
hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)

plt.imshow(cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB))
plt.axis('off')
plt.title("HSV Image")
plt.show()
```




```
h, s, v = cv2.split(hsv)

plt.figure(figsize=(12,4))

plt.subplot(1,3,1)
plt.imshow(h, cmap='gray')
plt.title("Hue")
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(s, cmap='gray')
plt.title("Saturation")
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(v, cmap='gray')
plt.title("Value")
plt.axis('off')

plt.show()
```




```
lab = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)

lab_rgb = cv2.cvtColor(lab, cv2.COLOR_LAB2RGB)

plt.imshow(lab_rgb)
plt.axis('off')
plt.title("LAB Image")
plt.show()
```




```
new_width = width // 2
new_height = height // 2

resized = cv2.resize(image, (new_width, new_height))

plt.imshow(cv2.cvtColor(resized, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.title("Resized Image")
plt.show()
```




```
rotated = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE)

plt.imshow(cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.title("Rotated Image")
plt.show()
```




```
horizontal_flip = cv2.flip(image, 1)

plt.imshow(cv2.cvtColor(horizontal_flip, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.title("Horizontal Flip")
plt.show()
```




```
vertical_flip = cv2.flip(image, 0)

plt.imshow(cv2.cvtColor(vertical_flip, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.title("Vertical Flip")
plt.show()
```




```
negative = 255 - image

plt.imshow(cv2.cvtColor(negative, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.title("Negative Image")
plt.show()
```




```
roi = image[100:400, 100:500]
```

```
plt.imshow(cv2.cvtColor(roi, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.title("Region of Interest (ROI)")
plt.show()
```




```
print("ROI Shape:", roi.shape)
print("ROI Height:", roi.shape[0])
print("ROI Width:", roi.shape[1])
print("ROI Channels:", roi.shape[2])
```

```
ROI Shape: (284, 400, 3)
ROI Height: 284
ROI Width: 400
ROI Channels: 3
```

```
plt.figure(figsize=(15,10))

plt.subplot(2,4,1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original")
plt.axis('off')

plt.subplot(2,4,2)
plt.imshow(gray, cmap='gray')
plt.title("Grayscale")
plt.axis('off')

plt.subplot(2,4,3)
```

```
plt.imshow(cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB))
plt.title("HSV")
plt.axis('off')

plt.subplot(2,4,4)
plt.imshow(lab_rgb)
plt.title("LAB")
plt.axis('off')

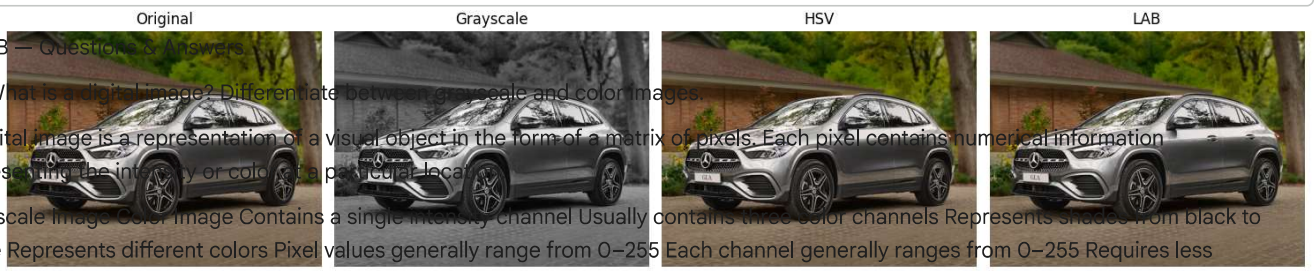
plt.subplot(2,4,5)
plt.imshow(cv2.cvtColor(resized, cv2.COLOR_BGR2RGB))
plt.title("Resized")
plt.axis('off')

plt.subplot(2,4,6)
plt.imshow(cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB))
plt.title("Rotated")
plt.axis('off')

plt.subplot(2,4,7)
plt.imshow(cv2.cvtColor(negative, cv2.COLOR_BGR2RGB))
plt.title("Negative")
plt.axis('off')

plt.subplot(2,4,8)
plt.imshow(cv2.cvtColor(roi, cv2.COLOR_BGR2RGB))
plt.title("ROI")
plt.axis('off')

plt.tight_layout()
plt.show()
```



Part B — Questions & Answers

Q1. What is a digital image? Differentiate between grayscale and color images.

A digital image is a representation of a visual object in the form of a matrix of pixels. Each pixel contains numerical information representing the intensity or color at a particular location.

Grayscale Image Color Image Contains a single intensity channel Usually contains three color channels Represents shades from black to white Represents different colors Pixel values generally range from 0–255 Each channel generally ranges from 0–255 Requires less storage Requires more storage Used when color information is unnecessary Used when color information is important

Q2. Explain the difference between RGB, BGR, HSV, and LAB color spaces.

Answer:



RGB: Represents an image using Red, Green and Blue channels. BGR: Similar to RGB, but the channel order is Blue, Green and Red. OpenCV normally uses BGR when reading images. HSV: Represents images using Hue, Saturation and Value. It is useful for color-based image processing. LAB: Separates lightness from color information and is designed to represent colors more independently of illumination.

Q3. What is the purpose of converting an image to grayscale before further processing?

Answer:

Grayscale conversion reduces a color image from three channels to one intensity channel. This reduces computational complexity and memory requirements. It is useful for operations such as edge detection, thresholding, contour detection and feature extraction where color information may not be necessary.

Q4. Explain the concept of image complement (negative image) and mention its practical applications.

Answer:

An image complement or negative image is produced by inverting the intensity of every pixel.

For an 8-bit image:

Negative pixel = 255 − Original pixel

For example, if a pixel has intensity 40:

255 − 40 = 215

Applications include:

Medical image analysis Enhancing certain image details Old photographic negative processing Document analysis Visualization of image features

Q5. Differentiate between image resizing, cropping, and scaling.

Answer:

Operation Meaning Resizing Changes the overall dimensions of an image Cropping Removes unwanted portions and selects a specific region Scaling Changes the size of an image by a specified scale factor

For example, reducing a 1000×1000 image to 500×500 is resizing/scaling, while selecting only a 300×300 portion from it is cropping.

Q6. What is a Region of Interest (ROI), and why is it important in computer vision?

Answer:

A Region of Interest (ROI) is a specific portion of an image selected for further processing or analysis